

Intelligent Anomaly Detection Method for Consumer Electronics Based on Feature-Driven Learning and Diffusion Model

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Abstract—The increasing complexity of consumer electronics and their dependence on continuous multivariate sensor data make intelligent anomaly detection a critical task. This paper proposes a novel anomaly detection framework, termed FD-Diffusion (Feature-Driven Diffusion Framework), tailored to time-series data generated by consumer electronic devices. The framework is composed of three key components. First, the Status-Aware Multimodal Encoder (SAME) extracts temporally aligned and semantically enriched representations from heterogeneous sensor streams by incorporating device status, behavioral context, and operational states. Second, the Condition-Aware Diffusion Model with Residual Refinement (CADM) leverages these embeddings to guide the reverse denoising trajectory of a diffusion process, ensuring that reconstructed sequences conform to normal behavior patterns. To enhance sensitivity to subtle deviations, CADM includes a residual refinement decoder with localized attention. Third, the Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM) fuses reconstruction loss, latent divergence, and semantic inconsistency into an interpretable and adaptive anomaly score, adjusted dynamically based on recent operational statistics. Experimental results on real-world device datasets demonstrate that FD-Diffusion outperforms existing methods in both detection accuracy and interpretability, offering a scalable solution for intelligent monitoring of consumer electronics under dynamic and diverse usage conditions.

Index Terms—Feature-driven learning, diffusion model, time-series anomaly detection, consumer electronics.

I. INTRODUCTION

THE EXPONENTIAL growth of smart consumer electronics has led to a sharp increase in real-time data generation, predominantly in the form of multivariate time-series originating from embedded sensors [1], [2]. These

sensors continuously monitor device states such as temperature, power consumption, signal strength, vibration, and hardware activity. Anomalies in this data can indicate software malfunctions, hardware degradation, or misuse, and early detection is essential for ensuring safety, reducing maintenance cost, and improving user experience [3], [4]. However, anomaly patterns are often subtle and device-specific, evolving gradually across temporal sequences and manifesting in complex, nonlinear ways. Traditional rule-based or statistical approaches fail to model such intricacies, especially when dealing with high-dimensional, asynchronous, and noisy data [5]. The increasing demand for intelligent and interpretable anomaly detection frameworks has pushed the boundary of research toward deep learning-based time-series modeling [6]. Techniques like RNNs [7], LSTMs [8], and TCNs [9] have been widely adopted for sequence analysis. They learn latent dynamics and temporal dependencies, enabling predictive capabilities far beyond manual feature engineering. In parallel, generative models such as VAEs [10] and GANs [11] have shown effectiveness in learning the probability distribution of normal operation states, enabling unsupervised anomaly detection without needing labeled faults. More recently, diffusion models have emerged as a powerful generative class, capable of progressively refining predictions through an iterative denoising process [12], [13].

Although deep learning has advanced time-series anomaly detection, current methods still suffer from critical shortcomings in real-world consumer electronics applications. Consumer devices generate temporally correlated but structurally diverse data streams, and simple early fusion or concatenation of features often leads to semantic conflicts or diluted signal importance [14]. Existing models frequently ignore inter-modal dynamics, resulting in suboptimal detection performance. Additionally, many architectures rely on fixed-window assumptions and exhibit poor generalization when data comes with variable lengths or contains asynchronous events [15]. Autoencoder-based models typically define anomaly scores using reconstruction loss; however, they may fail to detect small-scale or context-sensitive deviations due to their deterministic decoders [16]. LSTM and GRU models improve temporal awareness but encounter gradient vanishing and overfitting when sequences grow long or irregular [17]. VAEs and GANs, while capable of modeling data distributions, often produce oversimplified or unstable outputs, especially when applied to high-noise or non-stationary

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time-series [18], [19]. Furthermore, most existing models use static anomaly thresholds—manually set or statistically derived—that are not adaptive to device behavior over time. This leads to high false alarm rates or missed detections when the system enters new operational states or experiences gradual drift [20]. Moreover, anomaly scoring mechanisms seldom account for semantic alignment between reconstructed behavior and the underlying device status, making the results hard to interpret and trust [21].

To address these issues, we propose a novel deep anomaly detection architecture, termed Feature-Driven Diffusion Framework (FD-Diffusion), specifically designed for modeling and identifying anomalies in multivariate time-series data generated by consumer electronics. This architecture is jointly constructed from three complementary components. First, the Status-Aware Multimodal Encoder (SAME) is designed to extract latent semantic representations from asynchronous, heterogeneous sensor inputs while embedding contextual device information such as operational state, time dependency, and communication status. Through modality-specific encoding paths and cross-modal attention fusion, it aligns and integrates time-series sequences into a shared semantic space. Second, the Condition-Aware Diffusion Model with Residual Refinement (CADM) utilizes the embeddings produced by SAME to guide the generative reconstruction process. Each step in the reverse diffusion is conditioned on device-aware semantics, ensuring reconstructed outputs remain behaviorally consistent. Furthermore, a residual refinement decoder with stacked residual blocks and localized attention enhances the ability to preserve fine-grained variations critical to subtle anomaly detection. Third, the Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM) combines reconstruction loss, latent divergence, and semantic deviation into a unified anomaly score. By incorporating recent operational statistics, this module dynamically adjusts thresholds to support adaptive and interpretable detection in real-world environments. Together, SAME, CADM, and MAASM form the complete FD-Diffusion framework, which offers a robust, scalable, and real-time solution for anomaly detection in intelligent consumer electronics.

To summarize, this paper makes the following key contributions:

- We propose FD-Diffusion, a novel feature-driven diffusion-based framework tailored for time-series anomaly detection in consumer electronics, integrating representation learning, conditional generative modeling, and adaptive scoring into a unified architecture.
- We design the SAME to extract temporally aligned and semantically contextualized features from asynchronous, heterogeneous sensor inputs, enabling robust behavior-aware encoding.
- We develop the CADM that guides the reverse diffusion generation using semantic conditioning and enhances local anomaly recovery through residual refinement.
- We introduce the MAASM to dynamically integrate reconstruction, latent, and semantic signals with adaptive thresholding, improving detection sensitivity and interpretability.

- We conduct extensive experiments on a real-world dataset of consumer electronics and demonstrate that FD-Diffusion outperforms state-of-the-art baselines in accuracy, precision, recall, and F1 score.

The remainder of this paper is organized as follows: Section II reviews related work in time-series anomaly detection. Section III presents the overall architecture of the proposed FD-Diffusion framework and details its three core components: SAME, CADM, and MAASM. Section IV describes the dataset, implementation setup, and experimental procedures. Section V reports the evaluation results, including comparisons with baseline methods and comprehensive ablation studies. Finally, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

Machine learning-based anomaly detection algorithms primarily rely on distance- and density-based methods to identify anomalies. Rathore and Harsha [22] proposed a K-Nearest Neighbors (KNN) anomaly detection approach by calculating the distance between each data point and its k-nearest neighbors, and labeling the top m data points with the largest distances as anomalies. Xu et al. [23] used density-based methods, specifically the Local Outlier Factor (LOF), to detect outliers by analyzing the local density of data regions—where a larger LOF value indicates that a point lies in a lower-density neighborhood, and is thus considered anomalous. Liu et al. [24] introduced a novel approach combining Principal Component Analysis with noise handling and Density-Based Spatial Clustering of Applications with Noise (PCA-DBSCAN). This method applies dimensionality reduction and spectral data clustering to identify and remove anomalies, and incorporates tunable parameters to improve clustering performance. Fenza et al. [25] applied the K-means clustering algorithm to multivariate time-series electricity consumption data from 370 users, assigning preliminary category labels to each user's consumption profile, which helps in detecting anomalous consumption patterns in later tasks. Djenouri et al. [26] utilized the K-means method to detect anomalies in urban traffic flow data by constructing a probabilistic database of traffic flow distributions based on spatiotemporal data, and applying anomaly detection to future traffic forecasts—effectively addressing detection challenges under skewed data distributions. While these methods offer more flexible and sophisticated alternatives to purely statistical approaches and have shown effectiveness in many applications, they largely depend on manually defined anomaly thresholds, are highly sensitive to assumptions about data distributions and parameter settings, and require practitioners to possess strong domain expertise to properly configure and tune the models. Moreover, they are computationally expensive, require substantial memory resources, and generally lack robust generalization capabilities. Most importantly, these algorithms focus primarily on modeling and predicting features at individual time points, without accounting for the temporal dependencies inherent in time-series data.

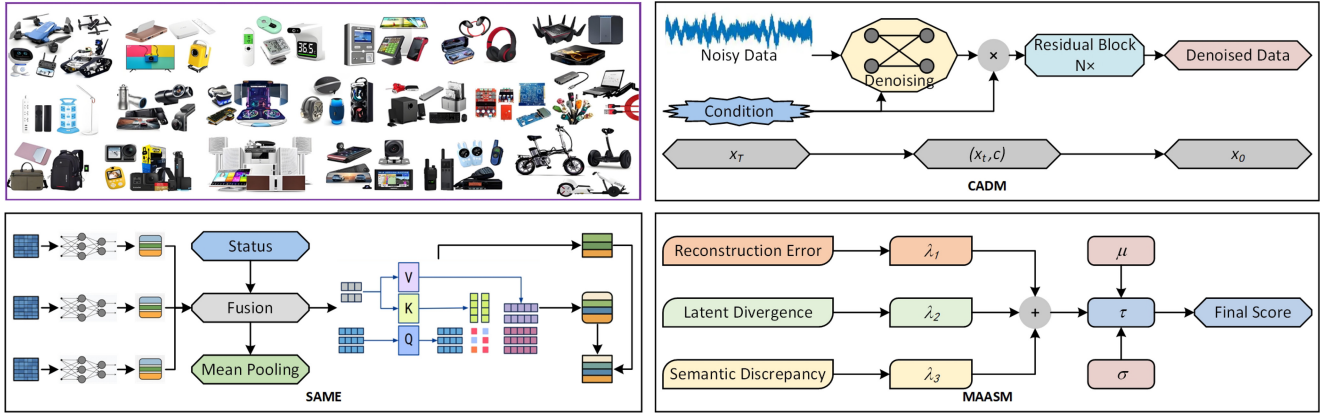


Fig. 1. Overall architecture of the proposed FD-Diffusion framework. It consists of three core components: SAME for multimodal encoding, CADM for condition-guided generation, and MAASM for adaptive anomaly scoring.

Deep learning-based anomaly detection models detect abnormal data by training deep generative models. Ergen and Kozat [27] studied unsupervised anomaly detection and introduced an LSTM-based neural network algorithm to identify product quality anomalies. Lindemann et al. [28] proposed a data-driven quality optimization method for system anomaly detection, where LSTM units were used to extract sequence features and an anomaly detection model was built using One-Class SVM (OCSVM). Zhang et al. [29] and Lu et al. [30] respectively employed Autoencoders (AE) and Denoising Autoencoders (DAE) to extract abstract features from vibration signals, overcoming the limitations of traditional handcrafted features. These models were trained on normal data using reconstruction error as the loss function. During testing, the reconstruction error served as the anomaly score, and values exceeding a set threshold were classified as anomalies. In addition, Chen et al. [31] improved performance by randomly varying the connection architectures of autoencoders, using different structural configurations and connection densities to reduce computational complexity. Combined with adaptive sampling, this approach aimed to enhance model diversity and reduce training time. Lin et al. [32] proposed a VAE-LSTM hybrid model for unsupervised time-series anomaly detection, integrating the representation learning power of the Variational Autoencoder (VAE) with the temporal modeling capability of LSTM. The VAE module captured structural patterns in local time windows, while the LSTM estimated long-term dependencies across sequences, forming a classical case of combining generative and predictive neural networks in anomaly detection. Compared to autoencoders, Generative Adversarial Networks (GANs) based on adversarial training can generate more detailed sample reconstructions. Inspired by GANs, Audibert et al. [33] proposed the USAD method for multivariate time-series anomaly detection. USAD features an autoencoder architecture with a two-phase adversarial training framework that improves model stability and reduces risks of collapse or non-convergence typical of GANs. Abul Bashar and Nayak [34] introduced TanoGAN for unsupervised anomaly detection in time-series data, modeling normal temporal behavior through adversarial training. Real-time time-series inputs are mapped to a latent space, then

reconstructed, and anomalies are detected based on the loss between real and reconstructed data.

III. METHOD

A. Overview of FD-Diffusion Architecture

The proposed Feature-Driven Diffusion Framework (FD-Diffusion) is an end-to-end architecture designed for anomaly detection in multivariate time-series data from consumer electronics. It integrates feature representation, generative reconstruction, and adaptive scoring to capture both temporal dependencies and semantic patterns in sensor data. The architecture is shown in Fig. 1. FD-Diffusion is composed of three core modules: the Status-Aware Multimodal Encoder (SAME), the Condition-Aware Diffusion Model with Residual Refinement (CADM), and the Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM). SAME extracts unified latent features from heterogeneous sensor streams while embedding operational context. CADM uses these features to guide a reverse diffusion process, generating behaviorally consistent reconstructions. A residual refinement decoder is incorporated to enhance sensitivity to subtle variations. MAASM fuses reconstruction error, latent divergence, and semantic deviation into a final anomaly score, which is evaluated through an adaptive thresholding strategy. Together, these three modules form a unified framework capable of accurate and interpretable anomaly detection in dynamic consumer device environments.

B. Status-Aware Multimodal Encoder

In real-world consumer electronics, devices are equipped with various sensors to monitor temperature, current, CPU usage, network signal strength, vibration, and more. These multivariate sensor streams form a high-dimensional and heterogeneous time-series signal that reflects the dynamic behavior of the device. However, these signals are not only multimodal in nature but are also inherently asynchronous, noisy, and behavior-dependent. An effective anomaly detection system must not only model intra-modality temporal dynamics but also integrate cross-modal semantics under the influence of time-varying operational status. To meet this requirement, we design the Status-Aware Multimodal Encoder (SAME).

The key challenge SAME addresses is the integration of asynchronous, semantically disjoint data streams into a coherent representation that is robust against device condition variations and sensor imbalance. Traditional early fusion approaches, such as direct concatenation or dimensional projection, are ill-suited to such scenarios as they ignore semantic misalignment and mutual dependence among modalities. Moreover, they lack the capability to incorporate explicit knowledge of the device's current state. Therefore, SAME introduces a hierarchical encoding strategy that consists of three major steps: modality-specific temporal encoding, status-aware feature augmentation, and attention-based cross-modal fusion.

Let $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M\}$ denote the set of M input modalities, where each $\mathbf{x}_m \in \mathbb{R}^{T \times d_m}$ is a time-series of length T and feature dimensionality d_m for modality m . These modalities may differ in sampling frequency or sensitivity to operational changes. Each modality is first processed by a temporal encoder $\phi_m(\cdot)$, which may take the form of a 1D convolutional encoder, a temporal convolutional network (TCN), or a lightweight Transformer, depending on the signal nature:

$$\mathbf{h}_m = \phi_m(\mathbf{x}_m), \quad \mathbf{h}_m \in \mathbb{R}^{T \times d_h} \quad (1)$$

To model context awareness, we define a status embedding vector $\mathbf{s} \in \mathbb{R}^{d_s}$ containing auxiliary behavior descriptors such as battery level, time-of-day encoding, operational state flag (e.g., charging vs. idle), or historical error count. These variables are conditionally independent of sensor values but critically affect the interpretation of those values. The status vector is broadcast across all time steps and concatenated to each time step's latent feature, enabling local time-aware semantic adjustment:

$$\tilde{\mathbf{h}}_m = [\mathbf{h}_m \| \mathbf{s} \cdot \mathbf{1}_T], \quad \tilde{\mathbf{h}}_m \in \mathbb{R}^{T \times (d_h + d_s)} \quad (2)$$

This augmented sequence is passed through a modality-specific multi-layer perceptron to project the features into a common latent space:

$$\mathbf{z}_m = \text{MLP}_m(\tilde{\mathbf{h}}_m), \quad \mathbf{z}_m \in \mathbb{R}^{T \times d_z} \quad (3)$$

At this point, each modality has a temporally resolved latent representation that is aware of device status. To unify semantics across modalities, we embed a cross-modal attention fusion mechanism as demonstrated in Fig. 2. This allows each modality to be compared and fused not only by similarity but also by relevance to the overall system context.

We first compute a global query vector $\mathbf{Q}$ representing overall device state by pooling across modalities:

$$\mathbf{Q} = \mathbf{W}_Q \cdot \text{MeanPool}([\mathbf{z}_1; \dots; \mathbf{z}_M]) \in \mathbb{R}^{1 \times d_z} \quad (4)$$

Each modality provides keys and values:

$$\mathbf{K}_m = \mathbf{W}_K \cdot \mathbf{z}_m, \quad \mathbf{V}_m = \mathbf{W}_V \cdot \mathbf{z}_m \quad (5)$$

Attention weights are computed as scaled dot-product similarity between the global query and modality-specific keys, followed by multiplication with values:

$$\mathbf{a}_m = \text{Softmax}\left(\frac{\mathbf{Q} \cdot \mathbf{K}_m^\top}{\sqrt{d_z}}\right) \cdot \mathbf{V}_m \quad (6)$$

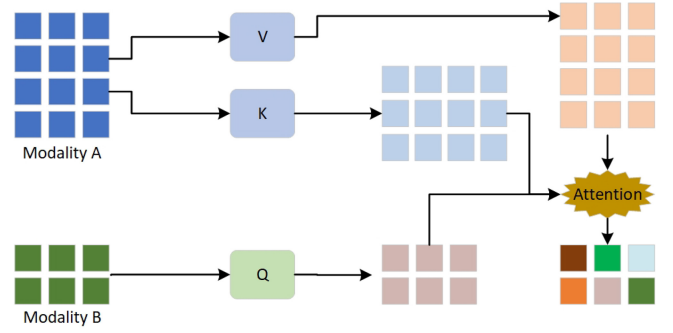


Fig. 2. Cross-modal attention mechanism.

We then average across the temporal axis to obtain modality-level embeddings $\bar{\mathbf{a}}_m$, and combine them using a learnable soft-attention fusion mechanism:

$$\alpha_m = \frac{\exp(\mathbf{w}_\alpha^\top \cdot \bar{\mathbf{a}}_m)}{\sum_{j=1}^M \exp(\mathbf{w}_\alpha^\top \cdot \bar{\mathbf{a}}_j)} \quad (7)$$

$$\mathbf{z}_{\text{fused}} = \sum_{m=1}^M \alpha_m \cdot \bar{\mathbf{a}}_m \quad (8)$$

The final fused representation $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{d_z}$ is used as the conditioning vector for the subsequent diffusion generation module.

Theoretical and practical significance: From a theoretical standpoint, SAME introduces a behavior-aware embedding mechanism that enriches classical temporal encoding with structured device state priors, improving representation quality under conditionally biased data distributions. Practically, the dynamic attention fusion allows the model to focus on the most informative modalities for a given operational context, improving robustness against missing or unreliable sensors. In the downstream anomaly detection process, such embeddings provide a strong prior to distinguish normal variations from behaviorally inconsistent or structurally implausible patterns.

Overall, the SAME module forms a foundational part of FD-Diffusion, enabling behavior-aware, temporally grounded, and semantically aligned representations across diverse sensor sources in complex electronic systems.

C. Condition-Aware Diffusion Model

To effectively detect anomalies in multivariate time-series data from consumer electronics, it is not sufficient to merely reconstruct inputs with low distortion. It is essential that the reconstruction process be guided by contextual behavior semantics, capable of reproducing the fine-grained, modality-specific dynamics that characterize normal device operation. The proposed Condition-Aware Diffusion Model with Residual Refinement (CADM) addresses this challenge by extending the standard denoising diffusion probabilistic model (DDPM) framework into a conditional and temporally refined generation process. CADM not only reconstructs input signals in a stepwise denoising fashion, but also injects behavioral context into the generative pathway and enhances local detail recovery through residual refinement.

In classical diffusion models, a forward process is defined that progressively corrupts the original data sample $\mathbf{x}_0 \in \mathbb{R}^{T \times d}$ with Gaussian noise across T steps. This stochastic process forms a latent trajectory $\{\mathbf{x}_t\}_{t=0}^T$ where t denotes the diffusion step:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad (9)$$

The set $\{\beta_t\}_{t=1}^T$ represents a fixed or learnable noise schedule, and the transition kernel ensures that the final state $\mathbf{x}_T$ approximates pure Gaussian noise. The goal of the generative model is to learn the reverse process that maps $\mathbf{x}_T$ back to $\mathbf{x}_0$ through a chain of denoising steps. In the unconditional setting, the model learns to predict the noise ϵ added at each step using a neural network ϵ_θ .

In CADM, this generative pathway is conditioned on a latent representation $\mathbf{c}$ produced by the encoder SAME, which embeds contextual behavior such as operational state, sensor interactions, and modality alignment. Thus, the reverse process becomes condition-aware:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t, \mathbf{c}), \Sigma_\theta(\mathbf{x}_t, t)) \quad (10)$$

To train the model, we adopt the simplified DDPM training objective, which minimizes the L_2 distance between the true noise and the predicted noise under a reparameterized noising schedule:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})\|_2^2] \quad (11)$$

where the input $\mathbf{x}_t$ is defined by:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}), \quad \bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i) \quad (12)$$

The conditional input is concatenated or cross-attended into each denoising block, allowing the network to tailor its reverse trajectory to the device-specific operational pattern.

To further enhance the temporal fidelity of the generated sequences, CADM incorporates a residual refinement module that improves the precision of local reconstructions. This component is particularly important in detecting soft or burst-like anomalies that manifest in small fluctuations across short temporal windows. In contrast to vanilla DDPM implementations where the output is decoded directly from the final denoising layer, CADM processes intermediate denoising outputs through multiple residual blocks before generating the final prediction. The residual connection is illustrated in Fig. 3. Each residual block enhances resolution and local consistency using depthwise temporal convolutions and gated attention mechanisms.

Let $\mathbf{h}^{(l)}$ denote the intermediate hidden representation at layer l . The residual refinement block updates it according to:

$$\mathbf{h}^{(l+1)} = \mathbf{h}^{(l)} + \mathcal{F}^{(l)}(\mathbf{h}^{(l)}, \mathbf{c}) \quad (13)$$

The residual transformation function is decomposed as:

$$\mathcal{F}^{(l)}(\mathbf{h}, \mathbf{c}) = \text{LayerNorm}(\mathbf{h}) + \text{DWConv1D}(\mathbf{u}) \quad (14)$$

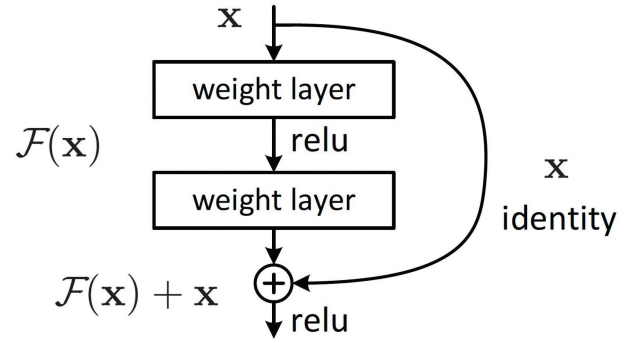


Fig. 3. The residual connection.

where the intermediate feature $\mathbf{u}$ is defined as:

$$\mathbf{u} = \text{MultiHeadAttn}(\mathbf{h}, \mathbf{h}, \mathbf{h}; \mathbf{c}) \quad (15)$$

This architecture enables the model to jointly consider global denoising context and short-range patterns that are essential for identifying small deviations such as jitter, micro-dropouts, or slight sensor drifts. The residual connection promotes gradient flow and stabilizes training, while attention layers dynamically adjust to the importance of various temporal regions under the given behavioral context.

At inference time, the model generates a reconstruction from noise by applying the learned denoising chain:

$$\hat{\mathbf{x}}_0 = \text{CADM-Denoise}(\mathbf{x}_T; \mathbf{c}) \quad (16)$$

The generated sequence $\hat{\mathbf{x}}_0$ is then compared to the observed sample $\mathbf{x}_0$ in the anomaly scoring module. Large discrepancies between these two reflect an inability of the model to conform the input to the learned distribution of normal behavior, which is indicative of anomalous states.

The design of CADM brings several benefits for the task of time-series anomaly detection in consumer electronics. The progressive denoising mechanism ensures that the model captures both coarse-grained and fine-grained structures throughout the reconstruction pathway. The condition-aware architecture allows the model to adaptively align reconstruction behavior with contextual semantics, which is essential for handling device heterogeneity and usage dynamics. The residual refinement decoder introduces an inductive bias toward local continuity, which is especially valuable when detecting short, sharp anomalies that would otherwise be averaged out in conventional architectures.

By combining structured probabilistic modeling with semantic guidance and localized reconstruction enhancement, CADM serves as a robust and expressive backbone for anomaly-oriented signal generation. Its outputs not only support direct anomaly scoring but also offer interpretable reconstruction pathways that reveal the dynamics of normal-versus-abnormal transitions over time.

D. Multi-Scale Adaptive Anomaly Scoring Mechanism

While generative reconstruction provides a valuable signal for identifying behavioral inconsistencies, effective anomaly detection requires a reliable scoring mechanism to quantify the

degree of abnormality for each input sequence. In real-world consumer electronics, anomalies may manifest through diverse temporal patterns and modality combinations—ranging from abrupt signal spikes to long-term drifts or minor structural distortions. These manifestations cannot be reliably captured by a single metric. To this end, we design the Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM), which combines three complementary perspectives of anomaly evaluation and integrates them under a dynamic thresholding framework to enable robust, interpretable, and context-aware anomaly detection.

Let $\mathbf{x}_0$ denote the original input sequence and $\hat{\mathbf{x}}_0$ the reconstruction output from the CADM module. The first scoring component is the reconstruction error score S_{rec} , which captures surface-level discrepancies between input and output:

$$S_{\text{rec}} = \frac{1}{T} \sum_{t=1}^T \left\| \mathbf{x}_0^{(t)} - \hat{\mathbf{x}}_0^{(t)} \right\|_2 \quad (17)$$

Although this score reflects visible distortions, it may fail to detect latent inconsistencies that arise from behaviorally implausible reconstructions. To address this, we introduce a latent divergence score S_{lat} , which measures the discrepancy between internal representations in the encoder:

$$S_{\text{lat}} = \left\| \phi(\mathbf{x}_0) - \phi(\hat{\mathbf{x}}_0) \right\|_2 \quad (18)$$

Here, $\phi(\cdot)$ represents a frozen instance of the SAME encoder used to extract semantic embeddings. This term provides a high-level semantic consistency signal, sensitive to misalignment in behavior patterns even if low-level reconstruction appears accurate.

To further capture semantic deviation from expected operational patterns, we define a condition-aware semantic discrepancy score S_{sem} between the input and its CADM-conditioned generation:

$$S_{\text{sem}} = \left\| f_{\text{sem}}(\hat{\mathbf{x}}_0, \mathbf{c}) - f_{\text{sem}}(\mathbf{x}_0, \mathbf{c}) \right\|_1 \quad (19)$$

where $f_{\text{sem}}(\cdot, \cdot)$ is a semantic decoder that projects a sequence under condition $\mathbf{c}$ into a behavior-specific feature space. Unlike S_{rec} or S_{lat} , this term emphasizes behavior-conditioned consistency.

The overall anomaly score S_{total} is computed as a weighted sum of the three components:

$$S_{\text{total}} = \lambda_1 S_{\text{rec}} + \lambda_2 S_{\text{lat}} + \lambda_3 S_{\text{sem}} \quad (20)$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and are either empirically chosen or learned using meta-optimization strategies on a validation set.

To adapt to different devices, conditions, and data regimes, we introduce a dynamic thresholding mechanism that computes the anomaly boundary based on recent score distributions. Given a sliding window of recent scores $\{S_1, S_2, \dots, S_N\}$, we define the adaptive threshold τ as:

$$\tau = \mu_S + \kappa \cdot \sigma_S \quad (21)$$

where μ_S and σ_S are the mean and standard deviation of the scores in the sliding window, and κ is a user-defined sensitivity factor. An input is classified as anomalous if:

$$S_{\text{total}} > \tau \quad (22)$$

This statistical method enables the detector to self-adjust to fluctuations in device usage patterns, operational states, or firmware updates without manual recalibration.

The design of MAASM reflects three important principles. First, it acknowledges the multi-faceted nature of anomalies in time-series signals by evaluating reconstruction quality, latent distribution alignment, and semantic conformity simultaneously. Second, it adapts to data drift and behavioral change via online statistical thresholding. Third, its modularity and interpretability make it suitable for integration into real-time consumer electronics applications, where anomaly explainability and reliability are essential.

By fusing low-level, mid-level, and high-level anomaly signals, MAASM significantly improves the precision and robustness of anomaly detection, particularly under weak anomaly conditions or non-stationary environments. It closes the loop from raw input to probabilistic reconstruction and finally to actionable decision-making, thereby completing the FD-Diffusion pipeline.

IV. EXPERIMENT

A. Dataset

To evaluate the effectiveness of the proposed FD-Diffusion framework, we conduct experiments on a real-world dataset collected from consumer electronic devices in practical usage scenarios. The dataset comprises multivariate time-series signals recorded from various internal sensors embedded in devices such as smart home terminals, wearable electronics, and portable communication modules. These sensors monitor parameters such as temperature, voltage, power consumption, CPU load, vibration, and wireless connectivity in real time. In total, the dataset consists of 903028 samples, each representing a short fixed-length segment of multivariate sensor readings. These samples reflect a wide range of device operation modes, including standby, active processing, charging, and communication phases. Manual inspection and logging records were used to identify both normal and anomalous behaviors, though anomaly labels are only used for evaluation purposes and not in model training. We divide the dataset into two non-overlapping subsets. Two-thirds of the samples are used as the training set, which contains only normal operation data and is used to learn the underlying behavioral distribution. The remaining one-third is used as the testing set, which includes both normal and anomalous sequences for evaluation. This split ensures a realistic setting where the model must detect unseen and unlabeled anomalies based solely on learned normal behavior during training.

B. Implementation Details

All experiments are implemented using PyTorch 1.13 and conducted on a workstation with an NVIDIA RTX 3090 GPU. Input time-series are segmented into windows of 128 time steps with a stride of 64, and each sensor channel is normalized using min-max scaling based on training data. The SAME encoder consists of three 1D convolutional layers followed by a cross-modal attention fusion module, with an output embedding size of 128. The CADM module is implemented

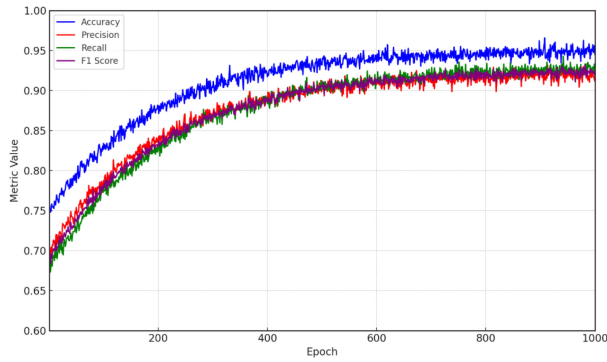


Fig. 4. Convergence curves of accuracy, precision, recall, and F1 score on the validation set during training.

as a 1D U-Net with residual refinement blocks; the number of diffusion steps is set to 1000, and the noise schedule follows a linear β_t from 10^{-4} to 0.02. For MAASM, the anomaly score is computed from reconstruction error, latent divergence, and semantic deviation, with fixed weights $\lambda_1 = 0.4$, $\lambda_2 = 0.3$, and $\lambda_3 = 0.3$, and dynamic thresholding is applied using a sliding window of 1000 samples and sensitivity coefficient $\kappa = 2.0$. All models are trained for up to 1000 epochs using the Adam optimizer (learning rate 1×10^{-4} , batch size 128) with early stopping and gradient clipping. During testing, a window is marked as anomalous if its score exceeds the adaptive threshold, and evaluation is performed using accuracy, precision, recall, and F1-score.

C. Convergence Evaluation

To verify the training stability and convergence behavior of the proposed FD-Diffusion framework, we track the evolution of four key performance metrics—accuracy, on the validation set throughout the training process. These metrics collectively reflect the model’s predictive quality and anomaly discrimination capability, and their consistent progression serves as evidence of effective and stable optimization. Fig. 4 presents the convergence curves of the four metrics over 1000 training epochs. All values are recorded on the validation set at the end of each epoch. As shown in the figure, the four metrics exhibit clear upward trends during the initial training phase and eventually stabilize, indicating that the model gradually learns representative normal behavior patterns and reaches a performance plateau without signs of overfitting or collapse. The bounded and monotonic convergence behavior across all metrics illustrates that FD-Diffusion is trainable under standard optimization settings. The smoothness of the curves also suggests that the design of the encoder, generative reconstruction module, and scoring mechanism contributes to a well-regularized learning process.

D. Comparison With Existing Methods

To validate the effectiveness of the FD-Diffusion framework, we compare it against existing anomaly detection methods. All models are trained and tested under the same dataset and evaluation protocol described in previous sections. The implementations follow publicly available repositories or

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Accuracy	Precision	Recall	F1 Score
THOC [35]	0.826	0.712	0.653	0.681
LSTM-VAE [36]	0.857	0.763	0.672	0.714
AnomTran [37]	0.842	0.749	0.641	0.690
TranAD [38]	0.861	0.770	0.699	0.732
TS2Vec [39]	0.893	0.811	0.777	0.793
DCdetector [40]	0.896	0.819	0.783	0.800
TimesNet [41]	0.901	0.826	0.789	0.807
MP-MixerNet [42]	0.918	0.845	0.817	0.831
MulGad [43]	0.912	0.834	0.801	0.817
FD-Diffusion (Ours)	0.941	0.881	0.863	0.872

reproduced settings from their original papers. The comparison is based on four standard metrics. Table I summarizes the experimental results. FD-Diffusion achieves the best performance across all evaluation metrics, significantly outperforming both classical and state-of-the-art baseline models. Specifically, FD-Diffusion obtains an accuracy of 0.941, a precision of 0.881, a recall of 0.863, and an F1 score of 0.872. Compared with the best-performing baseline method MP-MixerNet, which achieves an F1 score of 0.831, our method improves by 4.1 percentage points. When compared to earlier transformer-based models such as TS2Vec (F1 score 0.793) and TimesNet (F1 score 0.807), the gains reach 7.9% and 6.5%, respectively. Traditional sequential models such as THOC and LSTM-VAE yield F1 scores of only 0.681 and 0.701, with recall values below 0.67, highlighting their limited capacity in modeling long-range dependencies and complex inter-sensor correlations. Transformer-based methods including TranAD (F1 score 0.732) and AnomTran (0.690) show improved learning over recurrent architectures but still struggle to capture subtle, context-aware behavioral anomalies. Recent multivariate anomaly detection models like MP-MixerNet and MulGad demonstrate relatively strong performance, both achieving F1 scores above 0.81. However, their effectiveness tends to degrade under highly asynchronous conditions or when sensor modalities exhibit distributional imbalance. In contrast, FD-Diffusion leverages a behavior-aware encoder, condition-guided diffusion reconstruction, and a multi-perspective anomaly scoring mechanism. These components jointly contribute to maintaining high detection precision (0.881) while significantly improving recall (0.863), thereby achieving a strong balance between false positives and false negatives. The observed numerical improvements validate the advantage of integrating status-aware feature learning with conditional generative modeling. Furthermore, the use of adaptive scoring in MAASM enhances robustness to operational variability, ensuring that the framework adapts well to real-world fluctuations in device states. These results confirm the generalization capability and high sensitivity of FD-Diffusion in detecting diverse anomalies across complex consumer electronics scenarios.

E. Ablation Study of Key Modules

To evaluate the individual contributions of the three core modules in FD-Diffusion framework: SAME, CADM,

TABLE II
ABLATION STUDY OF SAME, CADM, AND MAASM

Configuration	Accuracy	Precision	Recall	F1 Score
Baseline	0.882	0.794	0.752	0.772
+ CADM	0.901	0.824	0.782	0.802
+ CADM + SAME	0.925	0.855	0.834	0.844
FD-Diffusion	0.941	0.881	0.863	0.872

and MAASM, we conduct an ablation study by incrementally removing or replacing each module and observing performance degradation. Specifically, we define the following variants: **Baseline**: A standard autoencoder-based reconstruction model without condition input or adaptive scoring. **Baseline + CADM**: Replaces the autoencoder with CADM, using only reconstruction loss for scoring. **Baseline + CADM + SAME**: Adds the SAME encoder to provide status-aware conditional guidance to the diffusion model. **FD-Diffusion**: Integrates SAME, CADM, and MAASM as the complete proposed architecture. The evaluation is conducted on the same test set using the standard metrics. Table II summarizes the results. As shown in Table II, introducing CADM in place of a traditional autoencoder improves F1 score from 0.772 to 0.802, representing a 3.0 percentage point gain. This demonstrates the advantage of the diffusion-based generative reconstruction mechanism, which enables better modeling of time-dependent structures and smoother recovery of normal behavior patterns. Further integrating SAME improves performance to 0.844 in F1 score, an additional gain of 4.2 points. This improvement is largely due to the enhanced semantic conditioning provided by SAME, which allows CADM to generate behaviorally consistent reconstructions under varying device states. The recall increases from 0.782 to 0.834, highlighting that SAME helps detect more subtle and context-specific anomalies that would otherwise be misclassified. Finally, the addition of MAASM as the scoring mechanism brings the full system to 0.872 F1 score, with the highest accuracy (0.941) and precision (0.881) observed among all configurations. Compared to the previous variant, this represents a further 2.8 percentage point gain in F1 score. MAASM’s integration of multi-perspective anomaly signals—reconstruction, latent, and semantic—combined with adaptive thresholding significantly reduces both false positives and false negatives, particularly under complex operational variations. Overall, this ablation study confirms that each module in FD-Diffusion provides distinct and complementary benefits. SAME improves semantic representation and context modeling, CADM offers flexible and high-fidelity generative capability, and MAASM enhances detection precision and robustness through multi-scale scoring. Their integration yields a powerful and generalizable anomaly detection framework.

F. Internal Ablation of the SAME Module

To assess the importance of each internal structure in the Status-Aware Multimodal Encoder (SAME), we conduct an ablation study focusing on three core components: status embedding, cross-modal attention, and modality-specific encoders. We evaluate four configurations: removing status

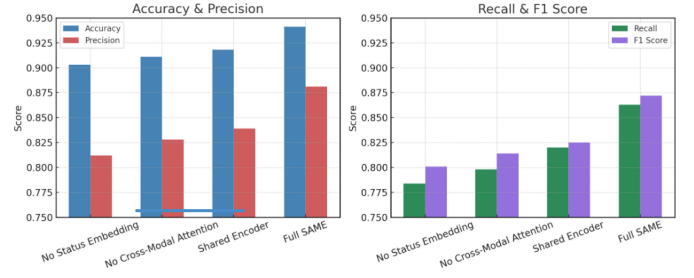


Fig. 5. Performance comparison of different SAME module configurations on four evaluation metrics.

embedding (No Status Embedding), removing cross-modal attention (No Cross-Modal Attention), replacing modality-specific encoders with a shared encoder (Shared Encoder), and using the full SAME module as deployed in FD-Diffusion (Full SAME). Fig. 5 illustrates the performance comparison across four evaluation metrics—accuracy, precision, recall, and F1 score—separated into two subplots for clarity. The complete performance values are derived from the validation set using consistent model settings across all variants. The results show clear performance differences among the four configurations. Removing the status embedding results in the largest performance drop, reducing F1 score from 0.872 to 0.801 and recall from 0.863 to 0.784. This 7.1 percentage point reduction in F1 score confirms that behavior-state representations such as communication patterns, temporal phases, and operational contexts are critical for accurate anomaly discrimination. Eliminating the cross-modal attention module yields an F1 score of 0.814, which is 5.8 points lower than the full model. In this case, the model loses the ability to explicitly capture inter-modality interactions, leading to reduced semantic consistency across sensor streams. Similarly, replacing modality-specific encoders with a shared encoder slightly improves over the previous two variants but still results in a 4.7-point drop in F1 score compared to the full SAME module. This indicates that shared encoders struggle to adapt to the heterogeneity of sensor modalities, while specialized encoders are better suited for learning domain-specific representations. In contrast, the full SAME configuration achieves the highest performance across all metrics: 0.941 accuracy, 0.881 precision, 0.863 recall, and 0.872 F1 score. These values are consistent with the overall FD-Diffusion framework, reaffirming that SAME plays a critical role in enabling condition-aware, semantically aligned, and behavior-sensitive representations that are essential for robust anomaly detection. In summary, this ablation study quantitatively confirms that each subcomponent of SAME contributes uniquely and substantially to the performance of the full system. The inclusion of status embeddings enables context-sensitive encoding, cross-modal attention improves semantic integration, and dedicated encoders enhance adaptability to diverse sensor types. Their combination is essential to support accurate and generalizable multivariate anomaly detection in consumer electronics.

G. Internal Ablation of the CADM Module

To further analyze the internal components of the Condition-Aware Diffusion Model with Residual Refinement

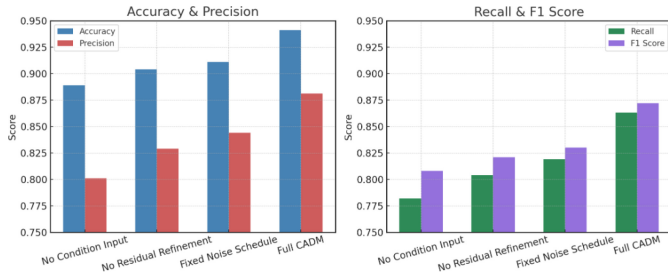


Fig. 6. Performance comparison of different CADM module configurations on four evaluation metrics.

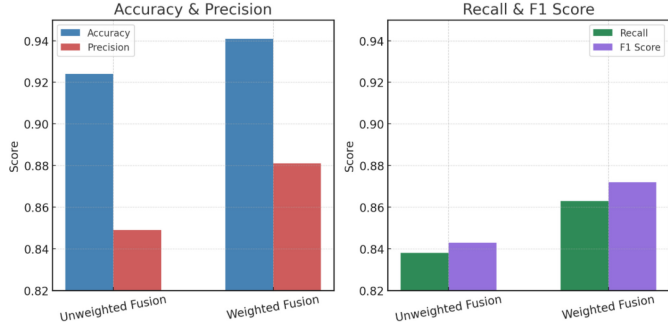


Fig. 7. Comparison between unweighted and weighted score fusion in MAASM across four evaluation metrics.

(CADM), we conduct an ablation study on three key structural mechanisms: the condition input from SAME, the residual refinement decoder, and the learnable noise schedule. Four configurations are evaluated: (1) removing condition input and performing unconditional denoising; (2) removing residual refinement and directly using the output of the last denoising layer; (3) replacing the learnable noise schedule with a fixed linear schedule; and (4) using the complete CADM as proposed. Fig. 6 presents the comparison of these configurations. The results show that each module within CADM plays an important role in the overall detection performance. Removing the condition input leads to the most significant performance degradation, with the F1 score decreasing from 0.872 to 0.808. This confirms that the semantic condition vector provided by SAME is crucial in guiding the generative process to match normal behavior patterns. The accuracy and precision also drop to 0.889 and 0.801 respectively, indicating reduced reliability and discrimination power. Removing the residual refinement mechanism results in a decrease in F1 score to 0.821. Although the base diffusion mechanism still functions, the lack of fine-scale residual modeling reduces the ability to capture localized anomalies. Similarly, using a fixed noise schedule leads to a F1 score of 0.830, suggesting that a learnable noise schedule offers better flexibility in adapting to various levels of temporal degradation and improves overall modeling fidelity. In contrast, the full CADM module achieves the best performance on all metrics, with an accuracy of 0.941, precision of 0.881, recall of 0.863, and F1 score of 0.872. These results validate the structural design of CADM and demonstrate that condition-aware guidance, residual refinement, and learnable diffusion noise schedules are all essential to achieving robust and fine-grained anomaly detection.

TABLE III
PERFORMANCE UNDER DIFFERENT COMPONENT IN MAASM

Configuration Variant	Accuracy	Precision	Recall	F1 Score
Remove S_{rec}	0.911	0.837	0.821	0.829
Remove S_{lat}	0.918	0.843	0.834	0.838
Remove S_{sem}	0.922	0.854	0.839	0.846
Full MAASM	0.941	0.881	0.863	0.872

H. Internal Ablation of the MAASM Module

To evaluate the effectiveness of the proposed Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM), we perform two sets of experiments. The first analyzes the contributions of each internal scoring component to the final performance. The second evaluates the effect of using weighted versus unweighted score fusion in the anomaly decision stage. We consider three individual components of the scoring mechanism: the reconstruction error score S_{rec} , the latent divergence score S_{lat} , and the semantic deviation score S_{sem} . We evaluate the performance when each component is removed from the scoring pipeline. The results are summarized in Table III. The results demonstrate that all three scoring components are essential to achieving strong anomaly detection performance. Removing the reconstruction score leads to the greatest F1 drop of 4.3 percentage points, confirming its role as a foundational metric for behavior reconstruction fidelity. Removing the semantic or latent-based scores also reduces performance, but to a lesser degree. These results validate the design of MAASM as a multi-perspective scoring framework. We further investigate the impact of using adaptive weights $\lambda_1, \lambda_2, \lambda_3$ to fuse the three component scores in the final anomaly scoring stage. Two configurations are compared: a simple unweighted average and a learned weighted fusion strategy. The comparison across four evaluation metrics is shown in Fig. 7. The experimental results show that using weighted fusion significantly improves performance across all metrics. The F1 score increases from 0.843 to 0.872, demonstrating that learned score importance enhances decision reliability. Precision also rises from 0.849 to 0.881, suggesting that adaptive fusion helps reduce false positives by emphasizing the most informative scoring signals in different scenarios. The combination of dynamic fusion and multi-perspective design makes MAASM both robust and interpretable for anomaly detection in complex environments.

V. CONCLUSION

In this paper, we proposed FD-Diffusion, a novel feature-driven diffusion-based framework for intelligent anomaly detection in consumer electronics. The framework consists of three key components: the Status-Aware Multimodal Encoder (SAME), which captures behavioral context and heterogeneous sensor dependencies; the Condition-Aware Diffusion Model (CADM), which reconstructs normal patterns through guided generation and residual refinement; and the Multi-Scale Adaptive Anomaly Scoring Mechanism (MAASM), which integrates multi-perspective evaluation metrics to quantify abnormality. Extensive experiments on a real-world consumer

electronics dataset demonstrated that FD-Diffusion significantly outperforms existing methods across multiple metrics. Ablation studies confirmed the individual effectiveness of each module, and internal component analyses validated the structural design of each sub-network. The proposed architecture not only achieves superior accuracy and generalization but also maintains interpretability and robustness under complex operational conditions.

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